Aditya Bidwai

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Education

University of Minnesota Twin Cities — *Minneapolis, MN, USA*
Master of Science (M.S.), Robotics — *Sep 2024 – Oct 2026*

Birla Institute of Technology and Science, Pilani (BITS-Pilani) — *Goa, India*
Bachelor of Engineering (B.E.), Electronics and Communication — *Aug 2018 – May 2022*

Experience

ACRES Lab, University of Minnesota Twin Cities — *Minneapolis, MN*
Research Assistant (Supervisor: Prof. Peng Wei) — *Oct 2026 – Present*

- Brought up two robot arms (YAM Pro, Seeed Studio): CAN/serial motor drivers, joint calibration, and ROS 2 control
- Developing robot manipulation techniques incorporating active perception for agricultural tasks (harvesting, pruning) under visual occlusion, benchmarking imitation learning policies (ACT, Diffusion Policy, VLA)

Komatsu Mining Corp — *Milwaukee, WI*
Robotics Software Intern (Automation) — *Jun 2025 – Aug 2025*

- Developed a C++ startup motor/encoder calibration routine for a robotic excavator, eliminating manual homing operations
- Built a human-intuitive C++ teleoperation interface for a robotic truck using a steering wheel and gas pedal (PX4/Arduino)
- Resolved OptiTrack tracking failures by tuning camera parameters (aperture/focus/calibration) for reliable autonomy testing

OptimalX Research Lab, University of Minnesota Twin Cities — *Minneapolis, MN*
Research Assistant — *Jan 2025 – Sep 2026*

- Deployed MARL policies on Crazyflie swarms; closed sim-to-real gaps via CrazySim SITL and PID rate-loop tuning
- Developed motion planning algorithms for UAV monitoring in energy-constrained environments (wildfires) using Mixed-Integer Linear Programming (MILP) -based optimization (Gurobi) and Markov Decision Process (MDP) methods
- Built an autonomous PX4/RPi UAV for GPS-denied indoor flight using ROS 2 offboard control and mocap + EKF2 **(video)**
- Developed a robotics research lab from scratch (truss design, OptiTrack camera system procurement) serving 500 students

MARMot Lab, National University of Singapore — *Singapore*
Research Engineer (Robotics) — *Jan 2022 – Aug 2024*

- Deployed a novel gaze-planning (heading optimization) algorithm for legged robot mapping with fixed field-of-view cameras on a real hexapod (ROS, RTAB-Map, RealSense) **(paper)**
- Developed an RL-based multi-robot task allocation method, 100x faster than optimization baselines **(ICRA '24 paper)**
- Deployed an inverse kinematics-based controller for omnidirectional locomotion for a hexapod in Python/ROS **(video)**

BITS Goa Mars Rover Team (Kratos), University Rover Challenge — *Goa, India*
Controls Team Lead — *Aug 2020 – Aug 2021*

- Designed path planning and tracking controllers (PID, Stanley, Pure Pursuit) for autonomous rover navigation
- Developed ROS/C++ drivers and software interfaces for system integration of sensors/motor drivers with an NVIDIA Jetson

Technical Skills

Programming	C, C++ (Modern C++), Python, MATLAB, Bash
Tools & Frameworks	Git, Linux, Docker, CMake, Bazel, **Robotics:** ROS, ROS2, MoveIt!, PX4, RViz, Gazebo, Isaac Sim
Hardware	Raspberry Pi, NUC, Nvidia Jetsons, Arduino, ESP32, STM32, RealSense cameras, Optitrack MoCap
Domain Interests	Optimal Control, Motion Planning, System Integration, Real World Deployment, Robotics Software

Selected Projects

Dynamic Visual SLAM for Crowded Indoor Environments

- Integrated NVIDIA SegFormer semantic segmentation into an ORB-SLAM3 pipeline to reject moving objects, achieving 0.077 m ATE and 0.073 m / 1.33° RPE in crowded real-world indoor environments

Clothbot: Dual-Arm Cloth Manipulation (poster, web)

- Developed a self-supervised value network policy for dynamic cloth unfolding on a dual UR5 setup, achieving 95% coverage on rectangular cloths and 87.68% on unseen garments with zero-shot sim-to-real transfer

Cooperative UAV-UGV Localization (code)

- Built a nonlinear Extended Kalman Filter (EKF) for cooperative UAV-UGV localization in MATLAB, validating estimator consistency through Monte Carlo NEES/NIS analysis on simulated and real data

Optimization-based 3D Motion Planning (code)

- Built a convex optimization framework (MATLAB, YALMIP/MOSEK) for collision-free 3D trajectory generation, modeling cuboidal and cylindrical obstacles as superellipse-based convex sets

Publications

AAMAS 2027 (under review): Cooperative Multi-Agent RL | **ICRA 2024** (2nd author): Multi-Robot Task Allocation via RL | **Frontiers 2023**: Legged Robot Manipulation Review | **ACM LANC 2022**: Flying Ad-Hoc Network Simulator